

Small's wooden plow

Scottish inventor James Small was a major contributor to the development of the plow. In the mid-1700s, he significantly redesigned the single-furrow horse plow and remodelled the mouldboard, which was now made of iron, making it stronger and more efficient at turning soil.

1950
3,400,000

1907
600

US 20th-century tractor usage

Tractors became more widespread as their design improved. They became smaller and mass production made them more affordable.

handle

moldboard turned soil to one side



THE AGRICULTURAL REVOLUTION

From the end of the 16th century in Europe, landowners began to take a more scientific approach to farming. This entailed following such practices as crop rotation to ensure that soil fertility was not depleted, using more efficient machinery such as improved plows, and cross-breeding livestock to avoid genetic weaknesses. While the success of these practices resulted in more food, it came at a cost: the need for farm laborers was greatly reduced, and land enclosures dispossessed many peasants who had nowhere to go.

1050–1300 The Middle Ages

The heavy plow, pulled first by teams of oxen then draft horses, changes the shape of fields, gives bigger yields, and greatly improves the way medieval Europe feeds itself.



Wooden plow

Late 18th century

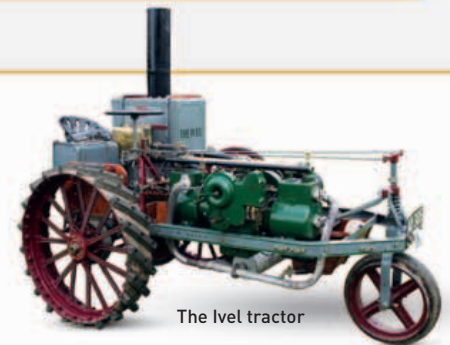
The cast-iron plow

The invention of first a cast-iron moldboard, then a cast-iron plowshare greatly improves plowing efficiency in both Old and New World agriculture.

1890s–early 1900s Gasoline-powered vehicles

The internal-combustion engine-powered agricultural tractor signals the end of horse-drawn farming.

By World War I, tractors become common parts of farm machinery in the US.



The Ivel tractor

1492

Columbian exchange
Columbus's discovery of the New World results in an unprecedented cultural exchange of animals and plants, including tobacco and corn.



Tobacco leaves

1800–40

The mechanical reaper

Although still horse-drawn, the mechanical reaper is the first step in the mechanization of farming, harvesting crops in much less time than could be achieved by hand.



McCormick's reaping machine

1970s

Move to sustainability

In a countermovement to the high-impact and environmentally damaging chemical practices of the mid-20th century, green and sustainable practices begin to be developed.

2000–present

Urban farming

As world populations rise, the search for new ways to feed more people increases. Vertical hydroponic farms in urban spaces have proved successful.